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UNIVERSITY
NAVI MUMBAI

Semester: IV

B. Tech. Computer Science and Engineering (Artificial Intelligence and Machine Learning)

38222401

CAC401

Artificial Intelligence

Date: 17-11-2025

Time: 02:00 PM TO 04:00 PM

Marks: 60

Q.1	Solve any Four	20 Marks	CO	BT
a)	Define Artificial Intelligence. Explain the characteristics of AI agents with an example of a vacuum-cleaner agent.	5	CO1	BT3
b)	Explain with diagram how heuristic search differs from uninformed search. Illustrate using best-first search on a route-finding problem.	5	CO2	BT4
c)	Differentiate between A* and AO* algorithms. Mention scenarios where AO* is preferred.	5	CO3	BT4
d)	Differentiate between Forward Chaining and Backward Chaining with a simple example.	5	CO4	BT4
e)	Consider two fuzzy sets $A = \{(1,0.2), (2,0.5), (3,0.8)\}$ and $B = \{(1,0.6), (2,0.3), (3,0.7)\}$. Compute $A \cap B$ and $A \cup B$.	5	CO5	BT5
f)	Write and explain the STRIPS representation for the action "Move (A, B, Table)" in the Blocks World domain.	5	CO6	BT5
Q.2	Solve any Four	20 Marks	CO	BT
a)	Formulate a Tic-Tac-Toe problem in state-space form showing all components of problem formulation.	5	CO1	BT3
b)	Illustrate Iterative Deepening Depth First Search with example.	5	CO2	BT3
c)	Define Simulated Annealing. Describe the role of the cooling schedule and temperature parameter.	5	CO3	BT3
d)	Write any three rules of First Order Logic (FOL) inference and give examples for each.	5	CO4	BT2
e)	Describe the working of a simple Fuzzy Controller used for room temperature control.	5	CO5	BT5
f)	Explain Partial-Order Planning with an example. How does it differ from total-order planning?	5	CO6	BT2



Q.3 Solve any Two

20 Marks

CO

BT

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| a) | Apply Iterated Hill Climbing to the function $f(x) = x^2 - 14x + 24$, starting from $x=0$, with random restarts in the range $[0,10]$. Show how local maxima/minima are avoided. | 10 | CO3 | BT5 |
| b) | Draw and explain the architecture of a Knowledge-Based Agent. How does it use the knowledge base, inference engine, and percepts to make decisions? Illustrate with an example of a simple reflex agent upgraded to a knowledge-based agent. | 10 | CO4 | BT3 |
| c) | Design a fuzzy inference system to control an automatic washing machine. Include fuzzification, rule base, inference, and defuzzification steps. | 10 | CO5 | BT5 |